# Laplace Transform

Bohaz

May 27th 2026

## I

The Laplace transform of $f(t)$ is defined as,

$$\mathcal{L}[f] = F(s) = \int_0^\infty f(t)e^{-st}\mathrm{d}t, \qquad f(t<0) = 0. \tag{1.1}$$

where $s$ is a complex number. $F(s)$ is defined for all $s$, that integral converges. However, when it is calculated, *it can be analytically continued to the whole s-plane.* Let,

$$f(t) = e^{at}h(t) \tag{1.2}$$

where $h$ is the Heaviside step function:

$$h(s) = \begin{cases} 0, & s < 0 \\ 1, & s > 0 \end{cases} \tag{1.3}$$

Then,

$$F(s) = \int_0^\infty e^{-(s-a)t}\mathrm{d}t = \frac{1}{s-a}, \qquad s > a. \tag{1.4}$$

The $F(s)$ has a simple pole at $s = a$ with residue $R(a) = 1$. The integral obviously diverges for $\mathrm{Re}(s) < a$. Nevertheless, $F(s)$ can be analytically continued to these parts of the $s$-plane. Let's find the inverse transform, let $s = x + iy$, so $F(s) = F(x, y)$ and,

$$F(s) = \int_0^\infty f(t)e^{-st}\mathrm{d}t = \int_{-\infty}^\infty e^{-xt}f(t)e^{-iyt}\mathrm{d}t \tag{1.5}$$

then by Fourier inversion $\mathcal{F}^{-1}$,

$$e^{-xt}f(t) = \frac{1}{2\pi}\int_{-\infty}^\infty F(x, y)e^{+iyt}\mathrm{d}t \tag{1.6}$$

thus,

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^\infty F(x, y)e^{+xt}e^{+iyt}\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^\infty F(x, y)e^{+st}\mathrm{d}t \tag{1.7}$$

then use $\mathrm{d}s = i\mathrm{d}y$, and changing the limit accordingly from $y$ to $s$, yields,

$$f(t) = \frac{1}{2\pi i}\int_{x-i\infty}^{x+i\infty} F(s)e^{+st}\mathrm{d}s \tag{1.8}$$

provided that $x$ lies on the right of all poles of $F(s)$ so that $F(x,y)$ is well-defined by the forward transform, so we may use Fourier inversion. Thus the inverse Laplace transform is given by the **Bromwich integral**,

$$\boxed{\mathcal{L}^{-1}[F] = f(t) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)e^{+st}\mathrm{d}s} \tag{1.9}$$

*where $c$ is to the right of poles of $F(s)$.*

Lets check this inversion using Eq. (1.4).

$$\mathcal{L}^{-1}[F] = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{e^{+st}}{s-a}\mathrm{d}s, \qquad c > a. \tag{1.10}$$

Consider the case of $t < 0$ first, we close the contour on the right using a large curved part $C_R$, since it does not enclose any poles, by Cauchy's theorem,

$$0 = \oint = \int_{c-i\infty}^{c+\infty} + \lim_{R\to\infty}\int_{C_R} \tag{1.11}$$

we use the following paramatrisation,

$$s = c + Re^{i\phi}, \qquad \phi : \frac{\pi}{2} \to -\frac{\pi}{2} \tag{1.12}$$

thus,

$$\int_{C_R} = \int_{\pi/2}^{-\pi/2} \frac{e^{ct}e^{Re^{i\phi}t}}{c + Re^{i\phi} - a} iRe^{i\phi}\mathrm{d}\phi \tag{1.13}$$

now take the limit of $R \to \infty$,

$$\lim_{R\to\infty}\int_{C_R} = \lim_{R\to\infty}\int_{\pi/2}^{-\pi/2} e^{ct}e^{Rt\cos\phi}e^{iRt\sin\phi}\mathrm{d}\phi \tag{1.14}$$

then notice that,

$$\cos\phi > 0, \qquad -\frac{\pi}{2} < \phi < \frac{\pi}{2} \tag{1.15}$$

thus,

$$\lim_{R\to\infty} e^{Rt\cos\phi} = 0, \qquad t < 0 \tag{1.16}$$

and,

$$\lim_{R\to\infty}\int_{C_R} = 0, \qquad t < 0 \tag{1.17}$$

therefore,

$$\int_{c-i\infty}^{c+i\infty} = \mathcal{L}^{-1}[F] = f(t) = 0, \qquad t < 0 \tag{1.18}$$

as anticipated.

Now consider the case of $t > 0$, we now need to close the contour on the left, by Cauchy residue theorem, we have,

$$2\pi i R(a) = \oint = \int_{c-i\infty}^{c+i\infty} + \lim_{R\to\infty}\int_{\pi/2}^{3\pi/2} \tag{1.19}$$

We can make the same paramatrisation, however this time we have,

$$\cos\phi < 0, \qquad \frac{\pi}{2} < \phi < \frac{3\pi}{2} \tag{1.20}$$

thus,

$$\lim_{R\to\infty} e^{Rt\cos\phi} = 0, \qquad t > 0 \tag{1.21}$$

and,

$$\lim_{R\to\infty} \int_{\pi/2}^{3\pi/2} = 0 \tag{1.22}$$

we also know the residue of this simple pole,

$$R(a) = e^{at} \tag{1.23}$$

finally we have,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} = \mathcal{L}^{-1}[F] = f(t) = e^{at}, \qquad t > 0. \tag{1.24}$$

overall,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{+st}}{s-a} = e^{at}h(t), \qquad c > a \tag{1.25}$$

thus this does indeed recover the original function. Notice that $F(s)$ was originally calculated for $s > a$ in Eq. (1.4). (i.e. on the right of pole), but for the $t > 0$, we closed the contour by going into the region where $s < a$, which is the region where the original integral for calculating $F(s)$ is not converged. We used used the analytic continuation of $1/(s-a)$ to $s < a$ to obtain the correct result, which is remarkable.